

HARM FROM FOREIGN BODY INGESTION IN ADULTS AND CHILDREN: A SYSTEMATIC REVIEW OF CASE REPORTS

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OBJECTIVES & BACKGROUND	TABLE SHOWING ANATOMICAL DISTRIBUTION OF HARM					
Foreign body ingestion is a common ED presentation, risks associated with batteries and magnets are well documented, clinical uncertainty remains for other objects Study aim: To evaluate harm patterns, anatomical locations of injury and the utility of imaging for low profile foreign bodies (excluding batteries/magnets)	Foreign body type	Oesophagus & Upper tract	Stomach	Small Intestine	Large Intestine	Other Locations
METHODOLOGY Design: Systematic review of case reports/series from EMBASE, Medline/PubMed, and CINAHL/Ebsco (inception to June 2025) Inclusion: Adults and children; intentional and accidental ingestion; clear outcome measures reported Exclusion: Batteries, magnets, poisons, and aspiration Dataset: 120 unique publications encompassing 138 individual cases.	Fishbone (25)	14 (56%)	1 (4%)	5 (20%)	3 (12%)	2 (8%)
	Dental Appliance (21)	10 (47.6%)	2 (9.5%)	6 (28.5%)	2 (9.5%)	1 (4.7%)
	Toothpick (17)	0	1 (5.8%)	6 (35.2%)	9 (52.9%)	1 (5.8%)
	Chicken Bone (10)	8 (80%)	0	1 (10%)	1 (10%)	0
	Blister pack (9)	3 (33.3%)	1 (11.1%)	5 (55.5%)	0	0
	Coin (8)	4 (50%)	2 (25%)	2 (25%)	0	0
	Needle (6)	2 (33.3%)	3 (50%)	0	1 (16.6%)	0
	Metal object (12)	5 (41.6%)	4 (33.3%)	2 (16.6%)	1 (8.3%)	1 (8.3%)
	Plastic (11)	5 (45.4%)	1 (9%)	5 (45.4%)	0	0
	Multiple objects (12)	2 (16.6%)	4 (33.3%)	2 (16.6%)	1 (8.3%)	2 (16.6%)
	Other objects (17)	7 (41.1%)	6 (35.2%)	0	1 (5.8%)	3 (17.6%)
	Total (147*)	60 (40.8%)	25 (17%)	34 (23.1%)	19 (12.9%)	9 (6.1%)
	*147 locations of harm arising from 138 case reports, as some cases involved more than 1 site					
RESULTS	CONCLUSIONS					
•Unique patterns of harm arise from relatively small objects, with characteristic patterns relating to age, object type, and site of complications •40.8% of harm occurred in the oesophagus and upper tract, with a substantial proportion occurring distally •Blister pack ingestion occurred almost exclusively in elderly patients (mean age 74), with harm predominantly at the small intestine (55.5%) •Toothpick harm was concentrated entirely distally with zero oesophageal cases, reflecting their tendency to pass proximally before causing injury •Dental appliances were the second most common cause of harm, spanning all anatomical levels •Chicken bones caused harm almost exclusively in the oesophagus (8 of 10 cases), contrasting with fish bones where distal complications predominated •CT remains the most sensitive modality yet misses radiolucent objects; serious harm can occur despite entirely normal scans •Fish bones visible in 75% of scans, toothpicks in only 25%, reflecting the radiolucent nature of wood	•Risk Stratification: Age + Object Type = Best Predictor of Risk •Harm frequently occurs after the object has passed beyond the stomach, and a well-appearing patient at the time of assessment does not preclude serious delayed complications •Discharging even a clinically well patient demands robust and explicit safety netting •Notably, drawing pins and safety pins were not associated with any cases of harm in this review, suggesting their design may confer a degree of inherent protection					

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